

M1 — 2-DOF model
validation (complete)

M3 — momentum
observer

M5 — motor-current
torque mapping

M7 — digging-force
model

M8 — field
validation
(CMPDI data)

M2 — load-torque
recovery (in progress)

M4 — measurement-
noise robustness

M6 — 4-DOF
extension

M9 — physics-informed
learning layer

—— complete

- - - in progress

—— planned

